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#include <stdio.h>

// ----- PGCD -----
long long pgcd(long long a, long long b) {
    if (a < 0) a = -a;
    if (b < 0) b = -b;
    while (b != 0) {
        long long r = a % b;
        a = b;
        b = r;
    }
    return a;
}

// ----- PPCM -----
long long ppcm(long long a, long long b) {
    long long d = pgcd(a, b);
    return (a / d) * b;
}

// ----- Euclide Étendu -----
long long euclide_etendu(long long a, long long b, long long *x, long long *y) {
    if (b == 0) {
        *x = (a >= 0) ? 1 : -1;
        *y = 0;
        return (a >= 0) ? a : -a;
    }
    long long x1, y1;
    long long d = euclide_etendu(b, a % b, &x1, &y1);
    *x = y1;
    *y = x1 - (a / b) * y1;
    return d;
}

// ----- Fonction 1 : Résolution équation ax + by = c -----
void equationLineaire() {
    long long a, b, c;
    printf("\n=== Resolution de l'equation a*x + b*y = c ===\n");

    printf("Entrer a : ");
    scanf("%lld", &a);

    printf("Entrer b : ");
    scanf("%lld", &b);

    printf("Entrer c : ");
    scanf("%lld", &c);

    long long d = pgcd(a, b);
    printf("\nPGCD(%lld, %lld) = %lld\n", a, b, d);
    printf("PPCM(%lld, %lld) = %lld\n", a, b, ppcm(a, b));

    if (c % d != 0) {
        printf("\n=> Pas de solution car %lld ne divise pas %lld.\n", d, c);
        return;
    }

    long long x0, y0;
    euclide_etendu(a, b, &x0, &y0);

    long long k = c / d;
    long long xp = x0 * k;
    long long yp = y0 * k;

    printf("\n=> Il existe une solution.\n");
    printf("Solution particuliere : x = %lld, y = %lld\n", xp, yp);

    printf("\nSolution generale :\n");

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        printf("x = %lld + %lld*t\n", xp, b / d);
        printf("y = %lld - %lld*t\n", yp, a / d);
    }

    // ----- Vérifier si nombre premier -----
    int estPremier(int x) {
        if (x < 2) return 0;
        for (int i = 2; (long long)i * i <= x; i++) {
            if (x % i == 0) return 0;
        }
        return 1;
    }

    // ----- Fonction 2 : Affichage nombres premiers -----
    void afficherPremiers() {
        int n;
        printf("\n=== Affichage des nombres premiers < n ===\n");
        printf("Entrer n : ");
        scanf("%d", &n);

        printf("Nombres premiers < %d :\n", n);

        for (int i = 2; i < n; i++) {
            if (estPremier(i))
                printf("%d\n", i);
        }
    }

    // ----- MAIN -----
    int main() {

        int choix;

        while (1) {
            printf("\n=== MENU PRINCIPAL ===\n");
            printf("1 - Resoudre ax + by = c (avec PGCD et PPCM)\n");
            printf("2 - Afficher les nombres premiers < n\n");
            printf("Votre choix : ");
            scanf("%d", &choix);

            if (choix == 1)
                equationLineaire();
            else if (choix == 2)
                afficherPremiers();
            else
                printf("Choix invalide.\n");
        }

        return 0;
    }

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